6.3700 Quiz 2 — short sheet: copy Side 1 and Side 2 onto one A4 sheet

SIDE 1 — L1–L11

Basics

$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, $P(B) = \sum_i P(A_i)P(B \mid A_i)$

Bayes: $P(A_i \mid B) = \frac{P(A_i)P(B|A_i)}{P(B)}$

$\frac{n!}{(n-k)!}$, $\binom{n}{k}$, $\frac{n!}{n_1!\cdots n_r!}$

$\operatorname{Var} X = E[X^2] - (EX)^2$, $\operatorname{Var}(aX+b) = a^2 \operatorname{Var} X$

indep: $E[XY] = EX\,EY$, $\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y$

Distributions (mean, var)

Bern(p)	$p,\ p(1-p)$	Bin(n,p)	$np,\ np(1-p)$
Geom(p)	$\frac{1}{p},\ \frac{1-p}{p^2}$	Pois(λ)	$\lambda,\ \lambda$
U$[a,b]$	$\frac{a+b}{2},\ \frac{(b-a)^2}{12}$	Exp(λ)	$\frac{1}{\lambda},\ \frac{1}{\lambda^2}$

Exp: $f = \lambda e^{-\lambda x}$, $P(X > t) = e^{-\lambda t}$ (memoryless)

Normal: $P(X \le x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$, $\Phi(-z) = 1 - \Phi(z)$

Joint / conditional

$f_X(x) = \int f_{X,Y}(x,y)\,dy$, $f_{X|Y} = \frac{f_{X,Y}}{f_Y}$

$f_{X|A}(x) = \frac{f_X(x)}{P(A)}$ on A, $E[X \mid A] = \int x f_{X|A}$

Bayes (any mix): posterior $\propto$ prior $\times$ likelihood, then normalize

$f_{X|Y}(x \mid y) = \frac{f_X(x)\, f_{Y|X}(y|x)}{f_Y(y)}$

Derived distributions

$F_Y(y) = P(g(X) \le y)$, then $f_Y = F_Y'$

$Y = aX + b$: $f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$

max: F^n; min: $1 - (1-F)^n$

$Z = X + Y$ indep: $f_Z(z) = \int f_X(x) f_Y(z-x)\,dx$

SIDE 2 — L12–L13

Covariance

$\operatorname{cov}(X,Y) = E[XY] - EX\,EY$

$\operatorname{cov}(aX+b, cY+d) = ac\operatorname{cov}(X,Y)$, bilinear

$\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{cov}(X,Y)$

$\operatorname{Var}\sum X_i = \sum \operatorname{Var} X_i + \sum_{i \ne j} \operatorname{cov}(X_i, X_j)$

$\rho = \frac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1,1]$; $|\rho| = 1$ iff linear

indep $\Rightarrow$ cov $= 0$ (not conversely: $X \sim U[-1,1]$, $Y = X^2$)

$\operatorname{cov}(I_A, I_B) = P(A \cap B) - P(A)P(B)$

indep: $\operatorname{Var}(XY) = E[X^2]E[Y^2] - (EX\,EY)^2$

Conditional expectation

$E[X \mid Y]$ is a r.v. (a function of Y)

$E\big[E[X \mid Y]\big] = E[X]$

$\operatorname{Var} X = E[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(E[X \mid Y])$

Random sum $Y = \sum_{i=1}^{N} X_i$:

$EY = EN\,EX$, $\operatorname{Var} Y = EN \operatorname{Var} X + (EX)^2 \operatorname{Var} N$

Random-bias coin, $Q \sim U[0,1]$: $\operatorname{Var} X = \frac{n}{6} + \frac{n^2}{12}$